MIL-A-43820A(GL)

7 March 1975
SUPERSEDING
MIL-A-43820(GL)
7 November 1972

MILITARY SPECIFICATION

AIR LOCK, (MUST)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an air lock fabricated from coated cloth and supported by aluminum bow frames. One end of the air lock is open, the other end has a detachable door assembly. The air lock, raised hemicylindrical in shape, has the following dimensions: Length 11 feet, width 5 feet, height 7 feet. Also covered is an erection kit and air lock light assembly (see 6.2, 6.7 and 6.8).

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

V-T-285	- Thread, Polyester
QQ-A-200/8	- Aluminum Alloy Bar, Rod, Shapes, Tube and Wire, Extruded, and Structural, Shapes, 6061
QQ-A-225/8	- Aluminum Alloy Bar, Rod, Wire, and Special Shapes: Rolled, Drawn, or Cold Finished, 6061
QQ-A-250/11	- Aluminum Alloy 6061, Plate and Sheet
QQ-A-601	- Aluminum Alloy Sand Castings
QQ-B-626	- Brass, Leaded and Non-Leaded, Rod, Shaped, Forgings, and Flat Products with Finished Edges (Bar and Strip)
QQ-P-416	- Plating, Cadmium (Electrodeposited)
QQ-S-698	- Steel, Sheet and Strip, Low-Carbon

FSC 5410

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TT-E-527	- Enamel, Alkyd, Lustreless
TT-E-485	- Enamel, Semi-Gloss, Rust-Inhibiting
TT-E-529	- Enamel, Alkyd, Semi-Gloss
TT-P-1757	- Primer Coating, Zinc Chromate, Low Moisture-Sensitivity
DDD-L-20	- Label; for Clothing, Equipage, and Tentage, (General Use)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard

MILITARY

MIL-P-501	- Pins, Tent, Metal
MIL-S-1734	- Slips, Tent Line
MIL-C-3432	- Cable and Wire, Electrical (Power and Control, Flexible and Extra-Flexible, 300 and 600 Volts)
MIL-A-3962	- Anchors, Ground, Arrowhead with Driving Equipment
MIL-W-4088	- Webbing, Textile, Woven Nylon
MIL-C-5541	- Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-C-6021	- Castings, Classification and Inspection of
MIL-W-6858	- Welding, Resistance, Aluminum, Magnesium, Non-Hardening Steels or Alloys, Nickel Alloys, Heat-Resisting Alloys, and Titanium Alloys, Spot and Seam
MIL-W-8604	- Welding of Aluminum Alloys, Process For
MIL-G-16491	- Grommet, Metallic
MIL-F-21840	- Fastener Tape, Hook and Pile, Synthetic
MIL-R-30500	- Rope, Polyester
MIL-C-43256	- Cord, Polyester, Solid Braid
MIL-C-43285	- Cloth, Coated (Chloroprene Base Coated, Chlorosulphonated Polyethylene Top Coated)

STANDARDS

FEDERAL

FED-STD-595	- Colors
FED-STD-751	- Stitches, Seams, and Stitchings

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MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U.S. Military Property
MIL-STD-403	- Preparation for and Installation of Rivets and Screws, Rocket and Missile Structure
MS 20392	- Pin, Straight, Headed-Drilled Shank
MS 20426	- Rivet, Solid, Countersunk 100 Degrees, Precision Head, Aluminum and Aluminum Alloy
MS 20604	- Rivet, Blind Nonstructural, Universal Head, Class 1
MS 20613	- Rivet, Solid-Universal, Head Steel, Carbon, and Steel, Corrosion-Resistant
MS 21209	- Insert, Screw Thread, Coarse and Fine, Screw Locking, Helical Coil, CRES
MS 24621	- Screw, Tapping-Thread Forming, Type B, Pan Head, Cross Recessed, Carbon Steel
MS 24665	- Pin, Cotter
MS 24693	- Screw, Machine, Flat, Countersunk Head, 100 Degrees, Cross Recessed, UNC-2A and UNF-2A (in./mm)
MS 25082	- Nut, Plain, Hexagon, Electrical-Thin
MS 27130	- Nut, Plain Blind Rivet-Flat and Countersunk Head, Open End
MS 27983	- Fasteners, Snap, Style 4
MS 33537	- Insert, Screw Thread, Helical Coil, Coarse and Fine Thread, Standard Dimensions for
MS 35207	- Screw, Machine - Pan Head, Cross Recessed, Carbon Steel, Cadmium Plated, UNF-2A (in./mm)
MS 35333	- Washer, Lock, Flat, Internal Tooth (in./mm)
MS 35340	- Washer, Lock - Helical Spring, Extra Duty
MS 51937	- Bolt, Eye - Shoulder
MS 51957	- Screw, Machine-Pan Head, Cross-Recessed, Corrosion Resisting Steel, Passivated, UNC-2A
MS 51960	- Screw, Machine Flat Countersunk Head, 82 Degrees, Cross, Recessed, Corrosion-Resisting Steel, Passivated, UNF-2A
MS 51971	- Nut, Plain, Hexagon-Steel, Corrosion-Resisting, 300 Series, Passivated, UNC-2B

AIR FORCE - NAVY AERONAUTICAL

AN 960-Washer, Flat

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DRAWINGS

ARMY NATICK LABORATORIES

5-4-1600	- Air Lock (MUST); Drawing List
5-4-1601	- Air Lock (MUST); Assembly Complete
5-4-1602	- Assembly and Details "A"
5-4-1603	- Assembly and Details "B"
5-4-1604	- Blackout Curtain Assembly
5-4-1605	- Light Assembly
5-4-1606	- Electrical Junction Box and Light Guard Assembly
5-4-1607	- Receptacle Bracket Assembly and Junction Box Cover
5-4-1608	- Bow Assembly
5-4-1332	- Shelter - Multi Purpose (MUST): Anti-Backdraft Valve Assembly
5-4-1333	- Damper and Grill Assemblies
5-4-1570	- Bow Spike Assembly

(Miniature copies of Drawings 5-4-1601, 5-4-1602, 5-4-1603, 5-4-1605 and 5-4-1608, identified as figures 1 through 5 respectively, are attached for information purposes only.)

(Copies of specifications, standards and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

Aerospace Material Specification (AMS)

AMS 2672	- Brazing Aluminum
AMS 3195	- Silicone Rubber-Sponge, Close Cell-Medium
AMS 3205	- Synthetic Rubber - Low Temperature Resistant, 45-55
AMS 3209	- Synthetic Rubber-Weather Resistant-Chloroprene Type, 65-75

(Application for copies should be addressed to Society of Automotive Engineers, Inc., 485 Lexington Avenue, New York, N.Y. 10017.)

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American Society for Testing and Materials (ASTM)

ASTM B 210 - Aluminum-Alloy Drawn Seamless Tubes

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race St., Philadelphia, PA 19103.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), the supplier shall furnish a sample of the item to be procured under this specification for first article inspection and approval (see 4.3 and 6.3).

3.2 Initial production testing sample. When specified (see 6.2), the supplier shall furnish a sample for initial production inspection and approval (see 4.4 and 6.4).

3.3 Inspection comparison testing sample. When specified (see 6.2), the supplier shall furnish a sample for comparison inspection and approval (see 4.5 and 6.5).

3.4 Materials and components. Materials and components shall conform to the specifications and drawings listed in section 2 and as specified herein.

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3.4.1 Cloth, polyester, coated. The coated polyester cloth shall conform to type I, class 1, 2 or 3 of MIL-C-43285, as specified on the applicable drawing, with the following exceptions:

a. That the adhesive used in preparing only the adherability test specimens, per MIL-C-43285, shall be as specified on Drawing 5-4-1601.

b. That the type I, class 1 coated fabric shall meet the 200°F adherability requirement of MIL-C-43285 with not more then 1/16 inch slippage.

3.4.2 Thread, polyester. All thread shall conform to type I, class 1, sub-class B of V-T-285. Dyed thread shall match olive drab shade S-1 (CA 66022) and shall show fastness to weathering equal to or better than the standard shade sample. When no standard shade sample is available, the dyed thread shall show good fastness to weathering. Thread sizes shall be as specified on the drawing.

3.4.3 Overcoating materials. Materials used for overcoating, 3.5.9, shall be air dry coatings meeting color and compositional requirements of the green top coating and pale green top coating of MIL-C-43285. When applied to a sample of coated cloth, as used in the shelter, and dried, the resulting sample shall meet the requirements in MIL-C-43285 for blocking resistance and flame resistance.

3.4.4 Dusting compound. The dusting compound shall be whiting, talc, or other finely divided mineral material. The compound shall not support mildew growth (see 4.8).

3.4.5 Adhesive. The adhesives shall be as specified on Drawing 5-4-1601. Containers of the adhesive base and the adhesive accelerator shall be marked with the manufacturer's batch number (a batch being that quantity of material manufactured at the same time using the same raw materials or subjected to some unit chemical or physical mixing process intended to make the final product homogeneous). Containers of the toluene are not required to be marked with a batch number and it is not required that the toluene be segregated and identified to a specific adhesive lot. Prior to use, each lot of adhesive shall be tested for dead load shear strength and dead load peel strength. The adhesive lot shall be used not later than 14 days after the date of acceptance (see 4.6.1.1).

3.4.5.1 Dead load shear strength. There shall be not more than 1/16 inch of slippage at 200°F when the adhesive is tested as specified in 4.9.1.

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3.4.5.2 Dead load peel strength. There shall be not more than 1/2 inch of peel at a temperature of 70° to 75°F when the adhesive is tested as specified in 4.9.2.

3.4.6 Anchoring components.

3.4.6.1 Pins, tent, metal. Metal tent pins shall conform to type II of MIL-P-501.

3.4.6.2 Anchor, ground. Ground anchors shall conform to type II, grade B of MIL-A-3962. Driving rods, driving heads and holding handles shall conform to MIL-A-3962.

3.4.6.3 Spike, bow. Bow spikes shall be as shown on drawing 5-4-1570, part no. 5-4-1570-1.

3.4.7 Alternate components. Components offered as equivalent to components specified hereinafter and on the referenced drawings as specific manufacturer's part number or equal shall be functionally equal, of equal or better quality, with the manufacturer's part number identified. The incorporation and inclusion of such a component in the design of the specified end product shall not require modification or change to any other specified component, and shall not reduce ease of maintenance to it or any other components, unless such modification or change is specifically approved by the contracting officer. Prior to manufacture of the preproduction first article, or if none is required, prior to commencing production, the contractor shall submit for the contracting officer's approval, a list identifying each proposed alternate component, together with proof that each listed component complies with requirements specified herein. The contracting officer, at his option, may require a physical sample of the proposed substitution. Approval of the submitted listing, together with necessary supporting data, authorizes the commencement of the first article or production manufacture, as applicable, and does not relieve the contractor of the responsibility that these components perform in accordance with specified requirements when incorporated into the end product.

3.5 Design and construction. Air locks shall conform to the drawings listed in section 2 and as specified herein.

3.5.1 Coated cloth components. All visible cloth surfaces of the air lock exterior shall be olive green and all visible inside surfaces shall be pale green.

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3.5.1.1 Body, end panel, flaps and blackout curtains. The air lock body, door assembly end panel, blackout curtains, electrical access cover flaps and slide fastener stringer flaps shall be fabricated from the class 3 coated cloth specified in 3.4.1.

3.5.1.2 Sod cloth, tent pin, bow support and reinforcing patches. The sod cloths, tent pin patches, outside reinforcing patches and bow support patches shall be fabricated from the class 1 coated cloth specified in 3.4.1.

3.5.1.3 Inside patches, reinforcing tape. The inside patches, light fixture patch assemblies, door assembly reinforcing tapes and inside tie patch assemblies shall be fabricated from the class 2 coated cloth specified in 3.4.1.

3.5.2 Stitching, machine. All stitching shall conform to type 301 of FED-STD-751, 5 to 7 stitches per inch.

3.5.2.1 Thread breaks. Thread breaks in stitching shall be overstitched not less than 1 inch at each break. Any thread break noted during inspection shall be repaired by overstitching the existing stitching, starting from a distance of one inch before the break, across the defective area, to a distance of one inch beyond the break. The ends of repair stitching are not required to be backstitched.

3.5.2.2 Stitching ends. The ends of stitching shall be overstitched not less than 1 inch except where ends are turned under in a hem, held down by other stitching or where stitching is performed automatically. When stitching is performed automatically on stitch patterns such as box and box with cross-stitch, three or more tying, overlapping or backstitches shall be used to secure the ends.

3.5.2.3 Skipped stitches and run-offs. Two or more consecutively skipped stitches or any run-off stitching shall be overstitched not less than 1 inch. Skipped stitches or run-offs noted during inspection shall be repaired as specified for thread breaks (3.5.2.1).

3.5.2.4 Polyester thread. The addition of any lubricant to the polyester thread prior to or during the sewing operation is prohibited. Water or air spraying for needle cooling is permissible.

3.5.3 Setting of rivets. Rivets shall be securely set and neatly headed.

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3.5.4 Setting of grommets. Holes punched to receive grommets shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommets shall be securely clinched without cutting the materials.

3.5.5 Setting of snap fasteners. If a hole is punched before inserting the male or female part of the fastener, it shall be smaller than the outside diameter of the fastener tube so that the tube must be forced through the hole. The fasteners shall be securely clinched.

3.5.6 Fusing of webbing ends. Cut ends of nylon webbing shall be fused and, after fusing, shall be free from frayed edges and loose fibers.

3.5.7 Fusing of polyester cord ends. Cut ends of braided cord shall be fused to prevent fraying.

3.5.8 Attachment of slide fastener stringers. Care shall be exercised to utilize proper tension during attachment of slide fastener stringers. Slide fastener scoops shall be evenly distributed throughout the entire length of component interface assemblies. The location of separating pins and separating pin retainers shall be as specified on the drawings. Uneven scoop distribution and improper location of pins or pin retainers is detrimental to the joining capabilities and interchangeability characteristics required. Care shall also be exercised to maintain specified distances between the outer edge of the chain and the stitching attaching the stringer tape to the cloth so that there will be no interference with the operation of the slider when mating the door assembly stringer to the body assembly stringer.

3.5.9 Cementing. Cementing of assemblies shall be in accordance with the following procedure:

a. Surface preparation. Surfaces to be bonded shall be clean and dry prior to application of the adhesive. Coated cloth surfaces to be bonded shall be thoroughly cleaned by rubbing with toluene-wetted clean cloths or pads, which may contain a mild, bonded abrasive of 80, or finer, grit size or may be of a coarse weave (as for scouring). If abrading is exercised, there shall be no scuffing or abrading of the polyester base cloth. Separate clean cloths and clean solvent shall be used for final cleaning wipes.

b. Adhesive mixing. The ratio of adhesive base to accelerator shall be as specified on drawing 5-4-1601. Prior to extracting quantities of the base and accelerator to be mixed, any settlement shall be uniformly redispursed. Containers for mixing, transfer and use shall be clean prior to each filling.

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Adhesive in the containers shall be covered during transit to work stations and during sustained periods of non-use to prevent loss of solvent by evaporation. After mixing, the adhesive shall be used only while spreadable enough to obtain an even coat.

c. Adhesive application. A minimum of two coats of mixed adhesive shall be evenly spread on the cleaned surfaces to be bonded. The first coat shall be dry to the touch before the application of the second coat. Prior to assembly, the adhesive coated surfaces shall not be allowed to be contaminated by a visible layer of dust or other foreign matter. Applied adhesive shall be free of heavy smears or excessively thick lap edges (resulting from insufficient spreading), thin starved areas, uncoated areas or other defects characteristic of improper application. The second coat shall be dried as appropriate for the assembly method to be used; i.e., to a tacky (not wet) aggressive grab condition for the two-way, dry stick method; or to a low-tack (dry condition) for two-way stick, solvent re-activation method.

d. Assembly. Assembly of components shall be made when the tack and grab of the adhesive coated surfaces are in a suitable range and to avoid a condition of either excess solvent in the adhesive layers or excessive drying or open time. Surfaces shall be joined in the locations specified on the drawings, applying adequate pressure and rolling or barring firmly to produce an adequate bond. Care shall be exercised to avoid finger contact in the adhesive coated areas along the edges of seams, patches, or other bonded parts. Precautions should be taken to prevent adhesive application, assembling or curing of an assembly at a temperature below 60°F. The assembled work shall not be moved too soon or in such a way as to result in edge lifting or a decrease in strength of bond until the adhesive has set. Surfaces shall be assembled on the same day that they were coated, or a third coat of adhesive shall be applied to each surface. When surfaces have been assembled in an incorrect location, the adhesive bond may be immediately separated and reset at the proper location. To facilitate separation, toluene may be applied at the bond line by squirt can or atomizer and the component then peeled away as the adhesive softens. The excess toluene solvent shall be allowed to evaporate before resetting the surfaces. When incorrectly located parts are separated and the adhesive is not in an active condition or has transferred to the other surface, the affected surface shall be given an additional coat of adhesive before re-assembling. Any adhesive residue shall not extend further than 1/2 inch beyond the edges of seams, splices, patches or other bonded parts. Any exposed adhesive within the 1/2 inch allowance shall be overcoated with material specified in 3.4.3. Overcoated areas shall be dusted with the dusting compound specified in 3.4.4. There shall be no unsightly smearing or spreading of adhesive or overcoating and care shall be exercised to insure that surrounding areas are kept clean.

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3.5.10 Light assembly. The light assembly shall be as shown on Drawing 5-4-1605. The length of cord required shall be as specified (see 6.2 and 6.8).

3.5.11 Welding. All welding shall be as specified on the drawings. Materials to be welded shall be clean and free from scale, corrosion, oil, water and other foreign matter. Welds shall be sound, smooth, free from flash and shall show good fusion. All scale and flux shall be removed from finished area.

3.5.12 Kit, erection. Unless otherwise specified (see 6.2 and 6.7), an erection kit, consisting of the following components, shall be furnished with each air lock assembly:

Item	Quantity	Reference
Pin, tent, metal, type II	8	3.4.6.1
Anchor, ground, arrowhead, type II, grade B	14	3.4.6.2
Driving rod, head and holding handle	1 each	3.4.6.2
Spike, bow	6	3.4.6.3

3.6 Finish of metal components.

3.6.1 Bow assemblies, light assemblies. All surfaces of the bow assemblies and air lock light assemblies to be painted shall be treated, primed and finished as specified on the drawings.

3.6.2 Anti-backdraft valve assembly. Prior to painting, all aluminum surfaces to be painted shall be cleaned and prepared in accordance with class 1A of MIL-C-5541. After surface preparation, all metal surfaces shall be coated with primer conforming to composition G or L, color T of TT-P-1757. When primer is dry, all exterior surfaces shall be coated with enamel conforming to TT-E-527, olive drab color No. 34087 of FED-STD-595. Interior surfaces shall be coated with enamel conforming to TT-E-529, pale green color No. 24554 of FED-STD-595.

3.6.3 Door assembly. All aluminum surfaces of the door assembly shall be cleaned and treated in accordance with class 1A of MIL-C-5541.

3.6.3.1 Doors. After surface treatment, all metal surfaces shall be coated with a primer conforming to TT-P-1757, composition G or L, color T. When the primer is dry, all exterior surfaces shall be coated with enamel conforming to TT-E-527, olive drab color No. 34087 of FED-STD-595. All interior surfaces shall be coated with enamel conforming to TT-E-527, pale green color No. 24554 of FED-STD-595.

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3.6.3.2 Frame. Those frame surfaces where coated cloth panels or tapes are to be cemented shall be primed with Bostik 1007P (see Drawing No. 5-4-1601).

3.6.4 Finish coats shall level out to an adherent, continuous and uniform film without runs, wrinkles, streaks or areas of no film. Any coating damaged during assembly or examination shall be touched up.

3.7 Marking.

3.7.1 Nomenclature. The application of the letters "US" and the nomenclature marking on the air lock shall conform to type IV, class 9 of DDD-L-20, in the size characters and locations shown on the drawings. Class 9 markings shall be clearly legible after subjection to accelerated weathering.

3.7.2 Identification labels. Identification labels shall conform to type II, class 6 of DDD-L-20 and shall be 3-1/2 inches by 5-1/2 inches minimum with a blank space of at least 1-1/4 inches provided on the bottom of the label for the Department of Defense inspector's stamp. Identification labels shall be stitched to the air lock in the location shown on the drawing.

3.7.3 Identification marking. Bow assemblies, air lock light assemblies and door assemblies shall be marked for identification in accordance with MIL-STD-130.

3.8 Workmanship. Cloth components shall be free of holes, cuts, or tears, thin or weak places caused by abrasion, exposed fabric, blisters, tunnels or delamination of coating, creases, wrinkles or coating missing. Webbing shall contain no frayed or scalloped edges. Cord and rope components shall be free of cuts, abrasions or breaks. Thread tension shall be maintained so that there will be no loose stitching and seam allowances shall be maintained with seams properly sewed so that no run-offs, pleats or open seams shall result. Care shall be taken in sewing so that there shall be no needle chews. All thread ends shall be trimmed to 1/2 inch or less. Metal components shall be free of burrs, sharp edges, corroded areas and shall not be broken or malformed. Cemented areas shall be free of puckers, loose edges, unadhered pockets or tunnels. There shall be no unsightly smearing or spreading of adhesive and care shall be exercised to insure that surrounding areas are kept clean and free from excess adhesive, stains or discoloration of coated cloth.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article (see 4.3).
b. Initial production inspection (see 4.4).
c. Initial inspection comparison (see 4.5).
d. Quality conformance inspection (see 4.6).

4.3 First article inspection. When a first article is required, the air lock assembly shall be examined for defects as specified in 4.6.3.1 through 4.6.3.6. The presence of any visual defect or any dimension not within the specified requirements shall be cause for rejection of the first article.

4.4 Initial production sample inspection. When an initial production sample is required, the air lock assembly shall be examined for defects as specified in 4.6.3.1 through 4.6.3.6. The presence of any visual defect or any dimension not within the specified requirements shall be cause for rejection of the initial production sample.

4.5 Comparison sample inspection. When a comparison sample is required, the air lock assembly shall be examined for the defects as specified in 4.6.3.1 through 4.6.3.6. The presence of any visual defect or any dimension not within the specified requirements shall be cause for rejection of the comparison sample.

4.6 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.6.1 Component and material inspection. The supplier shall insure that all components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards unless excluded, amended, modified or qualified in this specification or applicable purchase document.

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4.6.1.1 Adhesive testing. Each lot of adhesive shall be tested as specified in 4.9.1 for dead load shear strength, and as specified in 4.9.2 for dead load peel strength. The inspection lot of adhesive shall consist of all containers of adhesive base from the same batch, and all containers of accelerator from the same batch. The quantities of base and accelerator shall be as deemed appropriate for production use within the prescribed 14 day usage limitation as specified in 3.4.5. The sample unit for the dead load shear test shall be one cloth lap seam and two determinations shall be made per sample lap seam. The sample unit for the dead load peel test shall be one cloth laminate and one determination shall be made per sample laminate. Any nonconformance to the requirements of 3.4.5.1 or 3.4.5.2 shall be cause for rejection of the representative lot of adhesive. Containers of the adhesive base and the accelerator for each acceptable lot of adhesive shall be marked with the 14 day usage date and with a lot identity.

4.6.2 Process examination. Examination shall be made of the following fabrication processes to establish conformance with specified requirements. Whenever nonconformance is noted, correction shall be made to the process and all items processed.

a. That the color of all visible exterior and interior surfaces or coated cloth components and assemblies are in compliance with 3.5.1.

b. That no lubrication is applied to the polyester thread prior to or during sewing operations for compliance with 3.5.2.4.

c. That holes punched to receive grommets are in compliance with 3.5.4.

d. That holes punched to receive snap fasteners are in compliance with 3.5.5.

e. That wiring of the light assembly is in compliance with the schematic wiring diagram shown on Drawing 5-4-1605.

f. That use of the adhesive is in accordance with cementing instructions specified in 3.5.9.

g. That welds and welding are in compliance with 3.5.11 and the drawings.

h. That bow assemblies and light assembly components are chemically treated, prior to priming, as specified on the drawings.

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i. That priming of bow assemblies and light assemblies is as specified on the drawings.

j. That anti-backdraft valve assembly components are chemically treated, prior to priming, for compliance with 3.6.2.

k. That priming of anti-backdraft valve assembly components is as specified in 3.6.2.

l. That door assembly components are chemically treated, prior to priming, for compliance with 3.6.3.

m. That priming of doors is as specified in 3.6.3.1.

n. That those frame surfaces where cementing to cloth components is to be effected are coated with the cement primer, for compliance with 3.6.3.2.

o. That the length of slide fastener stringers, the number of scoops per stringer, the locations of stringer pins or retainers and the distribution of scoops are as specified herein and on the drawings.

p. That tent pin patches and tie patches are constructed as specified on the drawings.

4.6.3 End item inspection. The inspection lot shall be all air lock assemblies offered for inspection at one time. The sample unit shall be one air lock assembly.

4.6.3.1 Visual examination of air lock body assemblies. The air lock body assemblies with doors shall be examined for defects in table I. The inspection level for this examination shall be II and the acceptable quality levels (AQL's), expressed in terms of defects per hundred units, shall be 15.0 major, 65.0 major and minor A defects combined and 100.0 total major, minor A and B defects combined.

TABLE I. Classification of defects for visual characteristics for the air lock body assembly

Examine	Defect	Classification Major	Minor A	Minor B
Cloth, coated	Hole cut or teat	X		
	Exposed fabric; delamination of coating	X		
	Crease or wrinkle; blister; tunnel; scratch or abrasion		X	
	Lump or heavy coated area			X

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TABLE I. Classification of defects for visual characteristics for the air lock body assembly (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Webbing	Frayed or scalloped edge			X
	End not fused			X
Cord	Cut or abrasion		X	
	End not fused			X
Rope components	Cut or break	X		
Hardware	Broken	X		
	Malformed:			
	- fails to perform intended function	X		
	- but will perform intended function		X	
	Corroded area			X
	Burr or sharp edge which may cause injury in handling or damage to fabric	X		
Grommets	Improperly set, two or more teeth exposed			X
	Clinched loosely, allowing grommet to rotate around hole			X
	Clinched excessively tight; i.e., cutting fabric	X		
	Insecurely clinched; i.e., grommet or washer becomes disengaged	X		
Cementing	Puckers; loose edges, unadhered pockets or tunnels		X	
	Excess adhesive in sags or lumps evident through outer layers of fabric		X	
	Exposed adhesive beyond the overcoated area:			
	- on exterior surface			X
	- on interior surface		X	

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TABLE I. Classification of defects for visual characteristics for the air lock body assembly (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Seams and stitching	Open seams:			
	- 1/2 inch or less			X
	- more than 1/2 inch, but not more than 1 inch		X	
	- more than 1 inch	X		
	NOTE: A seam shall be classified as open when one or more stitches are broken, or when one or more skipped or runoff stitches occur. On double-stitched seams, a seam shall be classified as open when either one or both side of a seam are open.			
	Thread breaks overstitched less than 1 inch		X	
	Needle chews resulting in cut, tear or hole	X		
	Skipped stitches or run-offs overstitched less than 1 inch (where two or more consecutively skipped stitches occur)			X
	NOTE: Skipped stitches and run-offs not overstitched shall be classified as open seams			
Seam type	Wrong seam type	X		
Stitch type	Wrong stitch type	X		
Stitch tension	Loose, resulting in an exposed bobbin or top thread for more than 6 inches in length			X
	Tight as evidenced by puckering on seams for more than 6 inches in length			X

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TABLE I. Classification of defects for visual characteristics for the air lock body assembly (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Stitches per inch	Less than minimum specified:			
	- one stitch			X
	- two or more stitches		X	
	More than maximum specified:			
	- one stitch			X
	- two or more stitches		X	
	NOTE: Variation in the number of stitches per inch, caused by the operator speeding up the machine and pulling the fabric in order to sew over heavy places or heavy seams or in turning corners, shall be classified as follows: a. Within the minor "B" defect classification - no defect b. Within the minor "A" defect classification - minor "B" defect			
Stitching margin	Larger than specified by more than 1/8 inch:			
	- for more than 1 inch, but not more than 4 inches in length			X
	- for more than 4 inches in length		X	
Stitching gage	Not as specified		X	
Stitching ends	Overstitched less than 1 inch (except where ends are held down by other stitching, turned under in a hem or where stitching is performed automatically)			X

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TABLE I. Classification of defects for visual characteristics for the air lock body assembly (cont'd)

Examine	Defect	Classification Major	Minor A	Minor B
Stitching ends (cont'd)	Secured with less than three tying, overlapping or backstitches when automatic stitching is performed			X
Component and assembly	Any component part omitted or not fabricated of applicable referenced material	X		
Finish of metal components	Not finished	X		
	Not adherent; e.g., blistered, peeled or flaky		X	
	Coating not continuous or uniform; i.e., not coated or area of no film			X
	Scratch, dig or abrasion exposing bare metal or primer			X
	Color not as specified		X	
Welding	Missing, incomplete, burnt through, cracked, porous, fractured or otherwise not fused	X		
	Not type specified, not smooth, flux and scale not removed			X
Marking; labels	Omitted, incorrect, illegible, misplaced or size characters not as specified			X
Cleanness	Grease or oil stains; thread ends not trimmed to 1/2 inch or less throughout			X

4.6.3.2 Dimensional examination of air lock assembly. The air lock body assemblies with doors shall be examined for defects in table II. The inspection level for this examination shall be S-3 and the AQL's expressed in terms of defects per hundred units, shall be 2.5 major and 40.0 total major and minor defects combined.

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TABLE II. Classification of defects for dimensional characteristics for the air lock body assembly

Examine	Defect	Classification Major	Classification Minor
Overall dimensions (unless otherwise specified)	Smaller than specified dimensions less applicable minus tolerance indicated on drawings	X	
	Larger than specified dimensions and applicable plus tolerance indicated on drawings		X
Slide fastener stringers	Length not within specified tolerance	X	
	Location of stringer separating pin not as specified	X	
	Location of retainer not as specified	X	
	Distance between end stop and separating pin or retainer greater than 1 inch	X	
Location dimensions (unless otherwise specified)	Not within specified tolerance		X

4.6.3.3 Visual examination of box assemblies. The bow assemblies shall be examined for defects in table III. The inspection level for this examination shall be II and the AQL's expressed in terms of defects per hundred units, shall be 2.5 major and 10.0 total major and minor defects combined.

TABLE III. Classification of defects for visual characteristics for the bow assembly

Examine	Defect	Classification Major	Classification Minor
Finish	Not finished	X	
	Not adherent; e.g., blistered, peeled or flaky		X
	Coating not continuous or uniform; i.e., not coated or area of no film		X
	Corroded area		X
	Scratch, dig or abrasion exposing bare metal or primer		X

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TABLE III. Classification of defects for visual characteristics for the bow assembly (cont'd)

Examine	Defect	Classification Major	Minor
Construction and workmanship	Component missing	X	
	Any component not fabricated of applicable referenced material	X	
	Split, puncture, burr, rough or sharp edge, deformed	X	
Rivets and riveting	Rivet missing	X	
	Not specified type, size		X
	Rivet not peened; i.e., can be removed by hand	X	
Marking	Missing; incomplete; illegible or incorrect		X

4.6.3.4 Dimensional examination of bow assemblies. The bow assemblies shall be examined for compliance with dimensions specified. Any dimension not within the specified drawing requirements shall be classified as a defect. The inspection level for this examination shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.6.3.5 Visual examination of light assemblies. The light assemblies shall be examined for defects in table IV. The inspection level for this examination shall be II and the AQL's expressed in terms of defects per hundred units, shall be 2.5 major and 10.0 major and minor defects combined.

TABLE IV. Classification of defects for visual characteristics for the light assembly

Examine	Defects	Classification Major	Minor
Finish	Not finished	X	
	Not adherent; e.g., blistered, peeled or flaky		X

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TABLE IV. Classification of defects for visual characteristics for the light assembly (cont'd)

Examine	Defect	Classification Major	Minor
Finish (cont'd)	Coating not continuous or uniform; i.e., not coated or area of no film		X
	Scratch, dig or abrasion exposing bare metal or primer		X
Construction and workmanship	Component missing	X	
	Any component not fabricated of applicable referenced material	X	
	Burr, sharp edge, deformed	X	
Marking	Missing; incomplete; illegible, or incorrect		X

4.6.3.6 Dimensional examination of light assemblies. The light assemblies shall be examined for compliance with dimensions specified. Any dimension not within the specified drawing requirements shall be classified as a defect. The inspection level for this examination shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.7 Examination of preparation for delivery requirements. An examination shall be made to determine that packaging, packing and marking are in compliance with section 5 requirements of this specification. Defects shall be scored in accordance with table V. The sample unit shall be one shipping container fully prepared for delivery. The lot shall be the number of shipping containers offered for inspection at one time. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

TABLE V. Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application.

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TABLE V. Examination of preparation for delivery (cont'd)

Examine	Defect
Materials	Any component missing; any component damaged.
Workmanship	Bow sections not secured; guy lines not individually coiled and secured against unwinding.

4.7.1 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-601 and PPP-B-636, examination for defects in closure, waterproofing and reinforcing shall be in accordance with those specifications.

4.8 Certificate of compliance. Where certificates fo compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.9 Tests.

4.9.1 Adhesive dead load shear strength.

4.9.1.1 Preparation of sample lap seam. Two rectangular swatches of Type I, Class 1, MIL-C-43285 coated fabric shall be cut, each approximately 8 inches by 10 inches with warp threads running parallel to the 10 inch dimension. The swatches shall be cut from areas of the coated fabric that are free of waviness or distortion of the weave. A quantity of the adhesive base and the accelerator, sufficient to fabricate the test samples, shall be selected from the adhesive lot. The two swatches shall be joined with the adhesive to form a lap seam wherein the chlorosulphonated polyethylene side of one swatch is adhered to the chloroprene side of the other. The seam shall be made by overlapping the 8 inch ends for a distance of 2 inches, with care exercised that the warp directions of the two swatches are aligned. The surfaces to be bonded shall be cleaned, adhesive shall be applied, and the assembly shall be made in the same manner used in production (see 3.6.9). Allow the seam to cure at 70° to 75°F. for seven days prior to testing. Adhesives may be tested in less than 7 days curing, and if found to meet the requirements, will be acceptable.

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4.9.1.2 Specimens. One-inch wide strips shall be cut from the sample lap seam prepared as specified in 4.9.1.1, to provide 1-inch wide by 18-inch long specimens with warp threads running parallel to the 18-inch length.

4.9.1.3 Procedure. One end of each specimen shall be anchored in a suitable grip. A load of 40 pounds shall be fastened to the other end of the specimen. The anchoring grip and the load application grip shall be of a type which precludes failure of the specimen in or near the grips. The grips shall not be applied any closer than 2 inches to the lap seam and shall hold the specimen uniformly across the 1-inch dimension without twist. A suitable gage mark (e.g., ball point line) shall be made at the lap seam edge prior to loading to assist in determining the amount, if any, of relative seam slippage. The two specimens shall be tested for four hours at 200°F. After removal of load, a second gage mark shall be made at the lap seam edge. The amount of slippage shall be reported as the distance measured to the nearest 1/16 inch between center lines of the gage marks for compliance with 3.4.5.1. Slippage greater than 1/16 inch shall constitute a failure.

4.9.2 Adhesive dead load peel strength.

4.9.2.1 Preparation of sample laminate. A sample laminate shall be prepared in the same way as described in 4.9.1.1 except that (a) the two pieces of cloth shall have adhesive applied over most of the surfaces of both pieces except for approximately two inches at one end and (b) the two pieces shall be assembled in a full overlap in lieu of the 2 inch lap.

4.9.2.2 Specimens. One-inch wide strips shall be cut from the laminate. The uncemented end will provide tabs for conduct of the peel test. The tabs shall be further separated by hand for at least one inch prior to test.

4.9.2.3 Procedure. The specimen shall be subjected to a one pound force (including weight of grip) in dead load peel for 10 minutes at a temperature of 70° to 75°F. The anchoring grip and force application grip shall hold the tab ends uniformly across the one-inch dimension and the force shall be applied uniformly without twist. Before application of the load, a suitable gage mark shall be made at the line of peel on one of the tabs. At the end of 10 minutes of test, the force shall be removed and a second gage mark made in the same way on the same tab. The distance between the marks shall be measured to the nearest 1/16 inch as the amount of peel for compliance with 3.4.5.2. Peel greater than 1/2 inch shall constitute a failure.

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5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. The light assembly shall be removed from the air lock. Each bow assembly shall be detached from the air lock body and disassembled into its two components. The fabric body shall be separated from the door assembly end panel.

5.1.1.2 Packaging. The fabric body shall be neatly and compactly folded onto and within the rectangular area of the floor assembly. The six bow sections shall be neatly bunched and secured with flat strapping or tape. Guy lines shall be individually coiled and secured against unwinding. The light assembly shall be packaged in a snug-fitting fiberboard box conforming to type CF (variety SW) or SF, class domestic, grade 175, style RSC of PPP-B-636.

5.1.2 Level C. Air locks shall be packaged to afford adequate protection against physical damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets this requirement.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Each air lock, packaged as specified in 5.1, shall be packed in a snug-fitting cleated plywood shipping container conforming to overseas type, type 2 load of PPP-B-601. Components shall be positioned with the door assembly at the bottom of the container. Closure and strapping shall be in accordance with the appendix of PPP-B-601.

5.2.2 Level B. Each air lock, packaged as specified in 5.1, shall be packed in a snug-fitting, cleated plywood shipping container conforming to domestic type, style A or B of PPP-B-601. Components shall be positioned with the door assembly at the bottom of the container.

5.2.3 Level C. Air locks, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

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5.3 Marking. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The air lock covered by this specification is intended to provide a protected entry to all major shelters (multi-purpose, expandables and inflatables) of the MUST system.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. When a first article is required (see 3.1, 4.3 and 6.3).
c. When an initial production testing sample is required (see 3.2.4.4 and 6.4).
d. When an inspection comparison testing sample is required (see 3.3, 4.5 and 6.5).
e. Whether the 6 foot length or 25 foot length of the light assembly cord is required (see 1.1, 3.5.10, 6.8 and Drawing 5-4-1605).
f. When an erection kit is not required (see 1.1, 3.5.12 and 6.7).
g. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 First article. When a first article is required it shall be inspected and approved under appropriate provisions of Armed Services Procurement Regulations ASPR 7-104.55. The first article should be a preproduction sample, initial production item or other specific item described under the definition of a first article in the ASPR. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the first article.

6.4 Initial production test. When an initial production test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the initial production test sample.

6.5 Inspection comparison test. When a comparison test sample is required, the contracting officer should include specific instructions in all procurement instruments regarding arrangements for inspection and approval of the comparison test sample.

6.6 Cementing of coated cloth. The following system has been used in preparing seams where chlorosulphonated polyethylene coated surfaces of coated cloths are to be adhered to chloroprene coated surfaces:

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a. Abrade, by hand, the chlorosulphonated polyethylene surface only to the point of dullness and uniform roughness. Norton Company's Lightning Metalite Cloth - Aluminum Oxide 80-J or 3-M Company's Three-M-ite-Elek-Tro-Cut Cloth J-wt Aluminum Oxide are satisfactory for use in abrading.

b. Wash the abraded chlorosulphonated polyethylene surface and the nonabraded chloroprene surface with clean cotton cloth swabs wet with toluene.

c. Allow each washed surface to dry at room temperature for 15-20 minutes.

d. Brush apply one coat of adhesive (see 3.4.5) to each washed surface.

e. Allow the adhesive to dry at room temperature for 15-20 minutes.

f. Brush apply a second coat of adhesive.

g. After 15-20 minutes drying, reactivate each adhesive with toluene.

h. Assemble immediately, smooth by hand, followed by the use of a steel roller.

i. Cure the cemented assembly a minimum of 24 hours before movement and a minimum of seven days before straining the cemented joint.

6.7 Air lock anchoring components are included in the Erection Kit for Shelter, Inflatable, with Air Lock and Connector, Corridor, Inflatable (MUST) and are not required when the Air Lock is supplied as a component of that shelter.

6.8 Light assembly, air lock. The light assembly with 6 foot cord length is for use when the air lock is to be supplied with the Inflatable Shelter, and the 25 foot cord length is for use when the air lock is to be supplied with the Expandable Shelter.

Custodian:

Army - GL

Review activities:

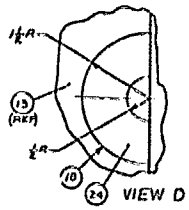
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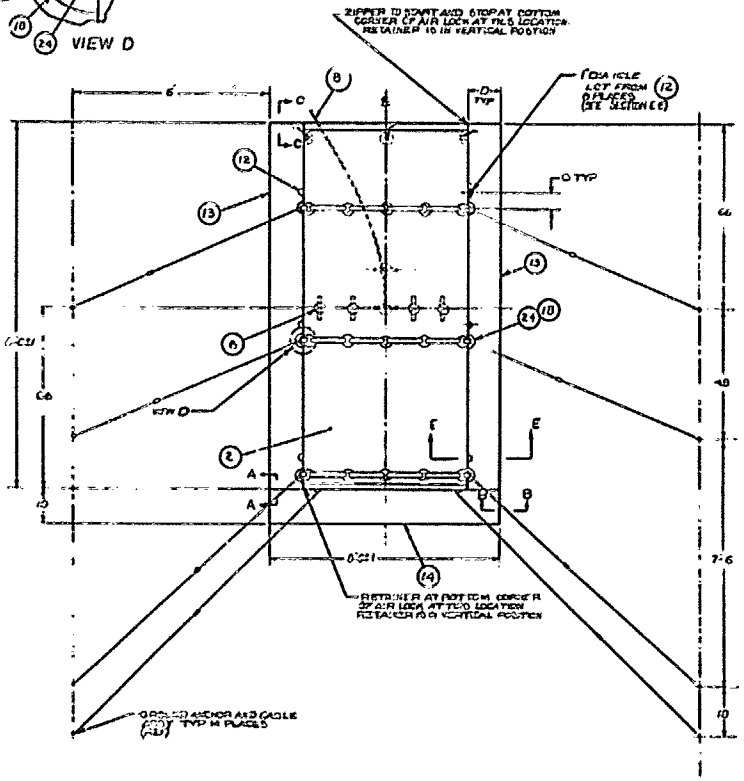
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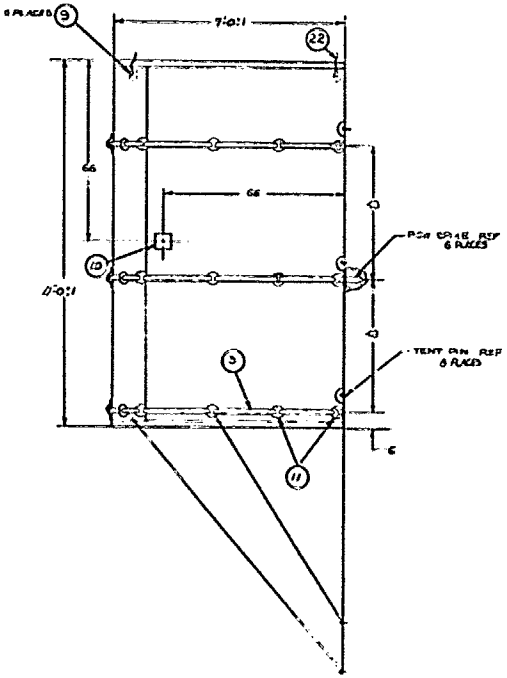
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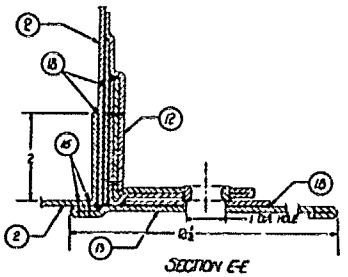
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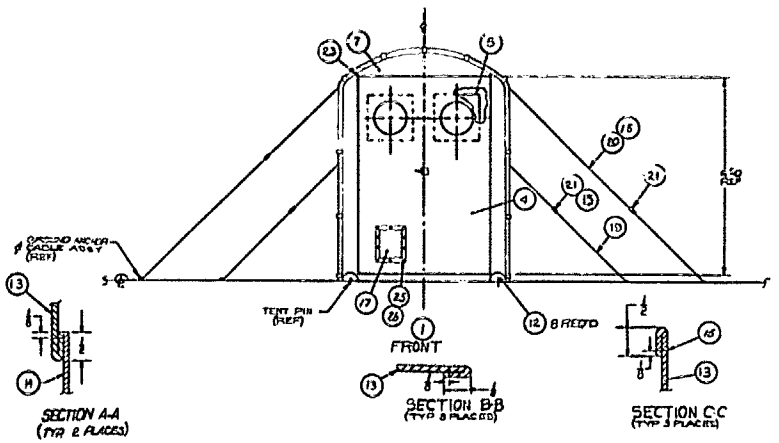
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NOTES:
1. COATED CLOTH SHALL CONFORM TO TYPE II OF MIL-C-43646 IN AIR LOCK AND WITH THE EXCEPTIONS CITED IN THIS SPECIFICATION.
2. ALL CUT ENDS OF WELDING FABRIC SHALL BE HEAT FUSED.
3. UNLESS OTHERWISE SPECIFIED, DIMENSIONAL TOLERANCE SHALL BE AS FOLLOWS:

LOCATING DIMENSION	TOLERANCE
3 TO BUT NOT INCLUDING	1/2
10 " " "	1
50 " " "	2
96 OR MORE	3

DETAIL ITEM DIMENSION	TOLERANCE
3 TO BUT NOT INCLUDING	1/8
10 " " "	1/4
36 OR MORE	1/2

4. ITEM 4 TO BE PLYCON INC P/N 1002-L OR APPROVED EQ. TYPE 1844B-NCO CLOTH
5. THE CONCENT SYSTEM TO BE USED EQUAL BY EITHER OF THE FOLLOWING:
   a. USM CHEMICAL CO SYSTEM NO. 10 59, 10 VOLUMES
      USM CHEMICAL CO GRADECOR NO. 5, 1 VOLUME
      TOLUENE 4 VOLUMES
   b. USM CHEMICAL CO SYSTEM NO. 1035, 16 VOLUMES
      USM CHEMICAL CO GRADECOR NO. 9, 2 VOLUMES
      TOLUENE 4 VOLUMES

NO. 1 THIS FIGURE IS A LOCATION COPY OF AIR MARCH LABORATORIES DRAWING AND IS FOR INFORMATION ONLY. THE DRAWING REFERENCE IN PARAGRAPH 3.1 GOVERNS.

REVISIONS			
LTR	DESCRIPTION	DATE	APPROVED
A	REVISED PER NOTE A	7 DEC 71	
B	ADDED 1 ITEM, REVISED 1 ITEM, DELETED 1 ITEM AS PER NOTE A	19 MAR 73	
C	ADDED 1 ITEM, REVISED 1 ITEM AS PER NOTE A	11 OCT 73	
D	ADDED 1 ITEM, REVISED 1 ITEM AS PER NOTE A	19 DEC 73	
E	REVISED PER NOTE A	7 MAY 74	

ITEM	QTY	DESCRIPTION
1	1	FABRIC, COATED, SEE NOTE 1
2	1	FABRIC, COATED, SEE NOTE 1
3	1	FABRIC, COATED, SEE NOTE 1
4	1	FABRIC, COATED, SEE NOTE 1
5	1	FABRIC, COATED, SEE NOTE 1
6	1	FABRIC, COATED, SEE NOTE 1
7	1	FABRIC, COATED, SEE NOTE 1
8	1	FABRIC, COATED, SEE NOTE 1
9	1	FABRIC, COATED, SEE NOTE 1
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20	1	FABRIC, COATED, SEE NOTE 1
21	1	FABRIC, COATED, SEE NOTE 1
22	1	FABRIC, COATED, SEE NOTE 1
23	1	FABRIC, COATED, SEE NOTE 1
24	1	FABRIC, COATED, SEE NOTE 1

DRAWN	PLYCON DIV
TRACED	
DESIGNED	
CHECKED	
APPROVED	
DATE	4 FEB 1969

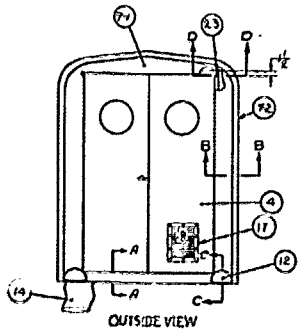
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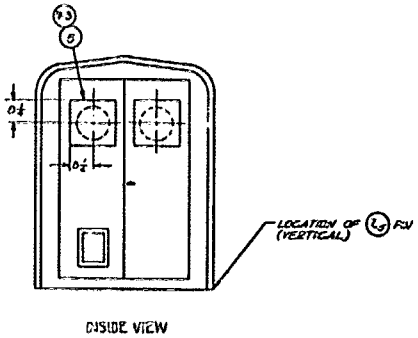
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ASSEMBLY COMPLETE

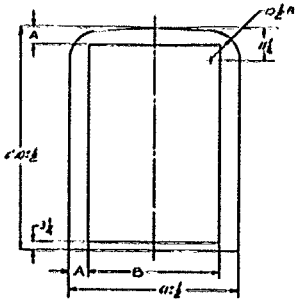
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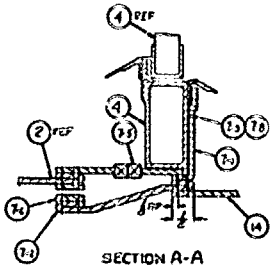
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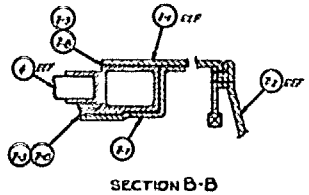
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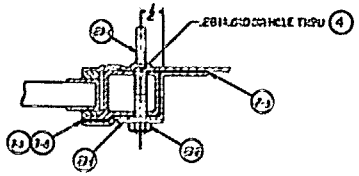


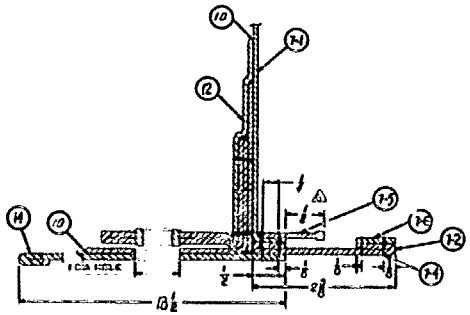


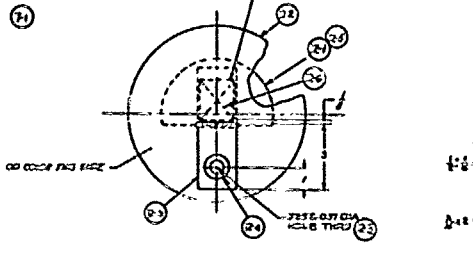


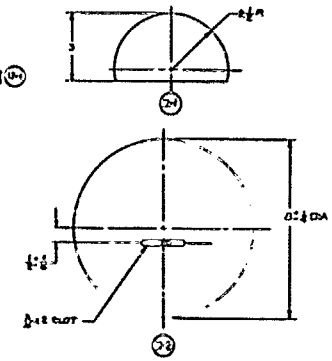












NOTE: DIMENSION B MUST BE SYMMETRICAL WITH EQUAL DIMENSION WITHIN 1/4 INCH.

NOTES:
4. UNLESS OTHERWISE SPECIFIED DIMENSIONAL TOLERANCES SHALL BE AS FOLLOWS:
   - LOCATING DIMENSIONS, LESS THAN 3 ——— ±1/8
   - 3 TO, BUT NOT INCLUDING 18 ——— ±1/4
   - 18 TO, BUT NOT INCLUDING 48 ——— ±1/2
   - 48 & OVER, BUT NOT EXCEEDING 120 ——— ±1
   - FOR ANGLES ——— ±1°
   - DETAIL ITEM DIMENSIONS; LESS THAN 3 ——— ±1/16
   - 3 TO, BUT NOT INCLUDING 15 ——— ±1/8
   - 15 TO, BUT NOT INCLUDING 36 ——— ±1/4
   - 36 OR MORE ——— ±1/2
5. DELETED
6. THE CONTROL SYSTEM TO BE USED SHALL BE EITHER OF THE FOLLOWING:
   a. USM CHEMICAL CO BOOSTIK NO. 1089, 16 VOLUMES
      USM CHEMICAL CO ACCELOR NO. 5, 1 VOLUME
      TOLUENE, 4 VOLUMES
   b. USM CHEMICAL CO BOOSTIK NO. 1005, 16 VOLUMES
      USM CHEMICAL CO ACCELOR NO. 9, 8 VOLUMES
      TOLUENE, 4 VOLUMES
7. THE FINISH SHALL CONFORM TO MIL-STD-171, EXCEPT THAT THE SCRIM LINING SHALL BE 1.5 OUNCES (DWG. 120 SHALL BE OBTAIN FILLED IN ACCORDANCE WITH QQ-P-416, CLASS 8, TYPE II.

NO. 2 THIS FIGURE IS A REDUCED COPY OF AN ARMY MATICK LABORATORIES DRAWING AND IS FOR INFORMATION ONLY.
FOR DRAWING REFERENCE SEE PARAGRAPH 3.1 COVERING.

ITEM LIST			
27			
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U. S. ARMY NATICK LABORATORIES

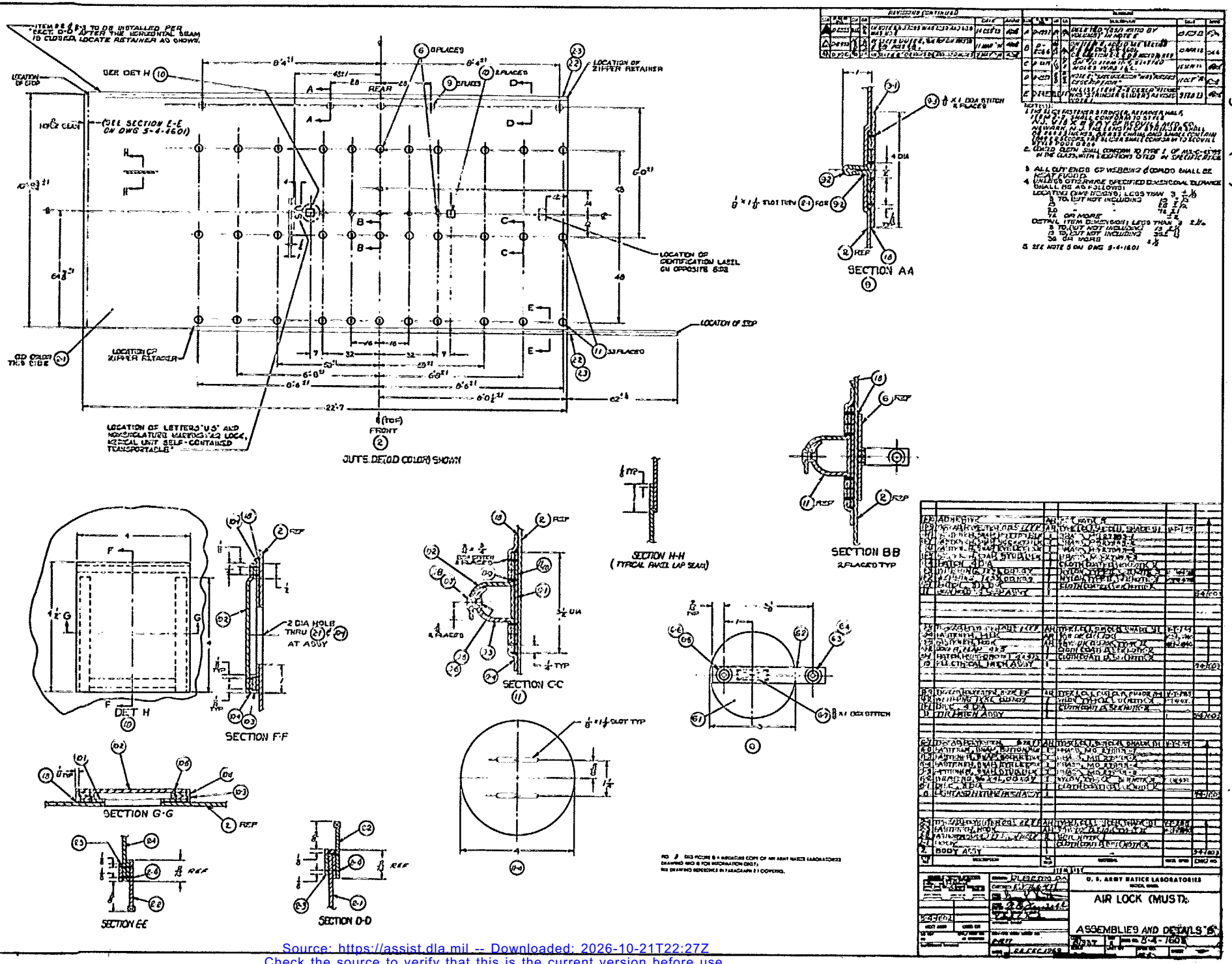
AIR LOCK, (MUST);

ASSEMBLIES AND DETAILS

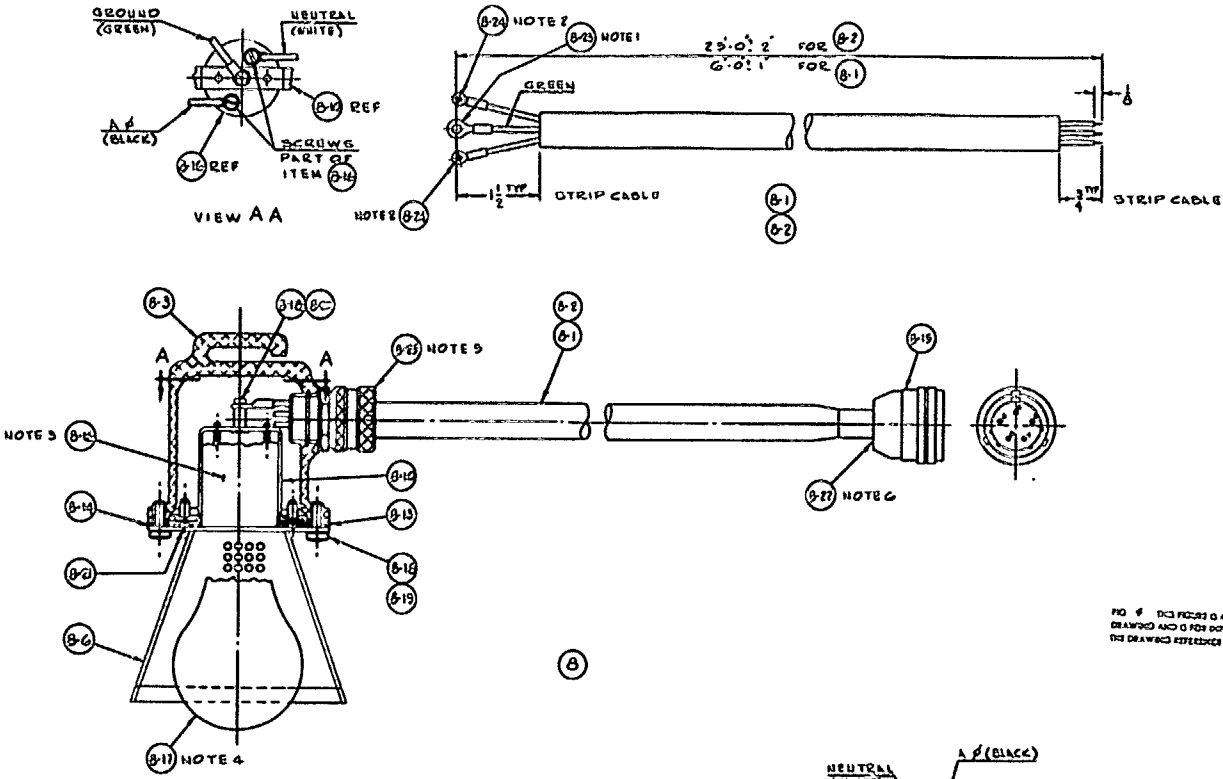
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29

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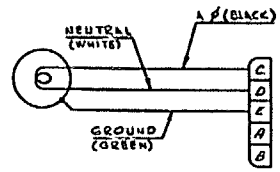


REVISIONS						
SYM	ZONE	LTR	DESCRIPTION	DATE	APPR
A	D-F853	S B	S WRG SCHEM FOR ITEM NO 8 WAS A, B, C, D LE SEQUENCE	17 OCT '72	[illegible]
A	D-B526	DCT	ADDED NOTE 8, ITEM DESCRIPTION AND [illegible]	[illegible]	[illegible]

FIG 8 THIS FIGURE IS A REPRODUCIBLE COPY OF AN ARMY NATICK LABORATORIES DRAWING AND IS FOR INFORMATION ONLY. THIS DRAWING REFERENCED IN PARAGRAPH 3.1 COVERING.

ITEM	DESCRIPTION	QTY PER ASSY	MATERIAL	MATL SPEC	NO.
A-25	CONNECTOR, CABLE	1	NOTE 8		
A-24	TERMINAL	2	COPPER (#10 WIRE - #6 STUD)		
A-23	TERMINAL	1	COPPER (#8-10 WIRE - #6 STUD)		
A-22	POTTING COMPOUND	AR	POLYSULFIDE (SEE NOTE 6) OR EQUIV		
A-21	SCR, MACH, EL-STOK NO 100°	6	CRES A-312 UNC-2A MS24693-C82		
A-20	SCR, MACH, PNH, CROSS REC	1	CRES A-312 UNC-2A MS19057-43		
A-19	SCR, MACH, PNH, CROSS REC	4	CRES A-312 UNC-2A MS19057-49		
A-18	WSHR LK, EL-INTL 100TH	8	STL CADPL MS35338-38		
A-17	LAMP	1	SEE NOTE 4		
A-16	SOCKET	1	SEE NOTE 3		
A-15	CONNECTOR, ASSY	1	SEE NOTES 8 AND *		5-4-1609
A-14	COVER, JUNCTION BOX	1			
A-13	GASKET	1			
A-12	RECEPTACLE BRACKET ASSY	1			5-4-1607
A-11	LIGHT GUARD ASSY	1			
A-10	ELECTRICAL JUNCTION BOX ASSY	1			5-4-1606
A-9	CABLE (3 WIRE)	1	TYPE CO-03449 F (3/16) 05C9	MIL-C-3432	
A-8	CABLE (3 WIRE)	1	TYPE CO-08MO F (3/16) 05C9	MIL-C-3432	
A	LIGHT ASSEMBLY	1			5-4-1608

ITEM LIST



WIRING SCHEMATIC FOR (8)

NOTES

1. AIRCRAFT MARINE PRODUCTS COMPANY PART NO 31888 OR EQUIV.
2. AIRCRAFT MARINE PRODUCTS COMPANY PART NO 36149 OR EQUIV.
3. CIRCLE "F" MFG CO PART NO 1813 OR EQUIV. REMOVE TOP AND SCRAP PRIOR TO USING IN ASSEMBLY.
4. GENERAL ELECTRIC COMPANY PART NO BCJ OR EQUIV.
5. KILLARK ELECTRIC MFG CO PART NO ZXB OR EQUIV.
6. COMPLETELY FILL AREA INSIDE OF POTTING BOOT WITH ITEM (A-22).
7. ITEM (A-1) IS FOR LIGHT ASSY USED WITH INFLATABLE SHELTER AND (A-2) IS USED WITH THE EXPANDABLE SHELTER.

NOTES: (CONTD)

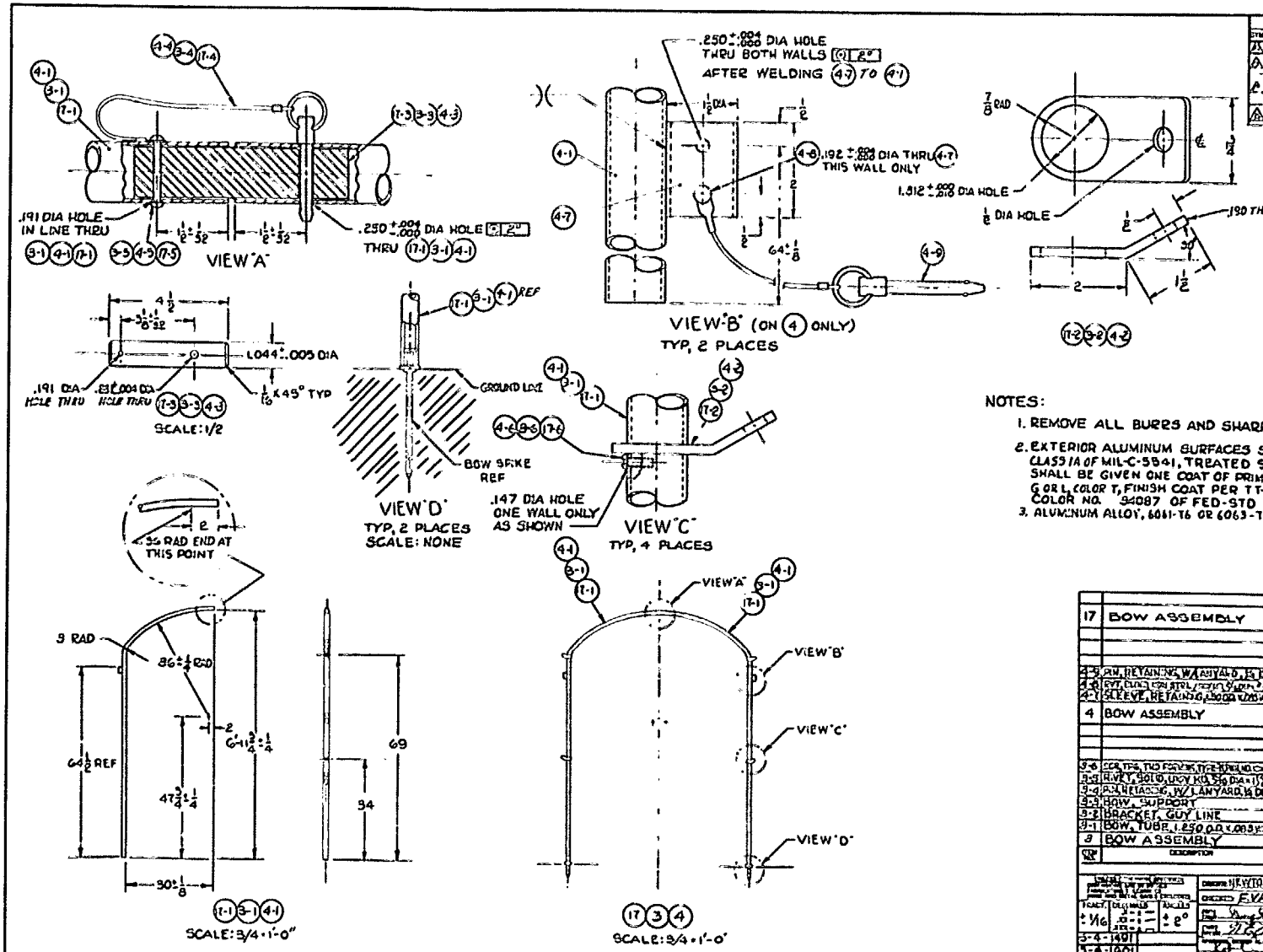
8. BENDIX, SCINTILLA DIV CO. PART NO. PT 01P18-8P OR EQUIV.
*8. FOR EXPANDABLE SHELTER ONLY, ITEM (A-15). USE BENDIX SCINTILLA DIV CO PART NO. PT 06P18-9P OR EQUIV.

DRAWN	A. J. FIELDEN	U. S. ARMY NATICK LABORATORIES, NATICK, MASS.	
CHECKED	R. BARKER	AIR LOCK (MUST)	
ENGR		LIGHT ASSEMBLY	
MFG		CAGE 81227	SIZE D
APPROVED		SCALE FULL	DWG NO. 5-4-1608
DATE	22 DEC 1969	UNIT WT	SHEET OF

MIL-A-4320A(GL)

31

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REVISIONS					
SYM	LTR	ZONE	DESCRIPTION	DATE	APPD
A	D-1137	G	REVISED MIL-A-8625 TO MIL-A-8625	16 DEC 71	G.E.S.
B	D-1937	G	IN LIST, ITEM 3-6 WAS 1/8 L, ON NOTE 2 COLOR NO WAS 34087	19 FEB 72	
C	D-1958	G	IN LIST, ITEM 3-1 MATL WAS 6061-T6 SEE NOTE 2 ABOVE SAME AS NOTE 2 IN 3-1. TOOL ROUND HOLE 3	1 OCT 72	
D	D3012	G	IN LIST 3-5 WAS ONE NO 2061SEC-24 ALSO REVISED NOTE 2	7 MAY 74	

NOTES:

1. REMOVE ALL BURRS AND SHARP EDGES.
2. EXTERIOR ALUMINUM SURFACES SHALL BE TREATED IN ACCORDANCE WITH CLASS 1A OF MIL-C-5541, TREATED SURFACES SHALL BE GIVEN ONE COAT OF PRIMER CONFORMING TO TT-P-1757, COMP G OR L, COLOR T, FINISH COAT PER TT-E-527, COLOR NO. 34087 OF FED-STD 595
3. ALUMINUM ALLOY, 6061-T6 OR 6063-T832 OF ASTM B-210.

ITEM	DESCRIPTION	REQ'D	MATERIAL	MATL SPEC	DWG NO.
17	BOW ASSEMBLY	1	ITEMS 17-1 THRU 17-6 ARE IDENT. TO 8-1 THRU 8-6		5-4-1608
4-5	PIN, RETAINING, W/ LANYARD, .5 DIA	2	AVDEL INC. 3500-5-6 OR EQUAL		
4-6	RIVET, CLOSED END, SOLID, 5/32 DIA	2	AL. MS20426AD5-4		
4-7	SLEEVE, RETAINING, .500 DIA	2	AL. SEE NOTES 2 & 3		
4	BOW ASSEMBLY	1	ITEMS 4-1 THRU 4-6 ARE IDENT TO ITEM 8-1 THRU 8-6		5-4-1608
3-6	CCR, TIE DOWN, W/ RING & HOOK	4	STL. CO FL. MS-24681-43		
3-5	RIVET, SOLID, UNIV HD, 5/32 DIA	1	AL. MS20470-D5N-24		
3-4	PIN, RETAINING, W/ LANYARD, .5 DIA	1	AVDEL INC. 3500-5-6 OR EQUAL		
3-3	BOW, SUPPORT	1	AL. 6061-T4 - SEE NOTE 2	QQA-200/6	
3-2	BRACKET, GUY LINE	4	AL. 6061-T6 - SEE NOTE 2	QQA-250/11	
3-1	BOW, TUBE, 1.250 O.D. X .083 WALL	2	AL. SEE NOTES 2 & 3		
3	BOW ASSEMBLY	1			5-4-1608

ITEM LIST

DESIGNED NEWTON S. DAVIS	U. S. ARMY NATICK LABORATORIES NATICK, MASS.		
CHECKED E.V. GENTI	AIR LOCK (MUST); BOW ASSEMBLY		
3-4-1971	CAGE 81337	SIZE B	DWG NO. 5-4-1608
5-4-1971	SCALE 1/1	WT	SHEET OF

FIG 5 THIS FIGURE IS A REDUCTURE COPY OF AN ARMY NATICK LABORATORIES DRAWING AND IS FOR INFORMATION ONLY; THE DRAWING REFERENCED IN PARAGRAPH 3.1 GOVERNS.

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